

Neuro-Fuzzy IDS with MRMR Feature Selection and Adaptive Ensemble Correction

1st Kunda Suresh Babu

Dept. of CSE

Narasaraopeta Engineering College

Narasaraopeta, India

sureshkunda546@gmail.com

2nd Pekala Veera Dhanush Kumar

Dept. of CSE

Narasaraopeta Engineering College

Narasaraopeta, India

pkveeragautham10@gmail.com

3rd Seggem Srinivas

Dept. of CSE

Narasaraopeta Engineering College

Narasaraopeta, India

sseggemsrinivas@gmail.com

4th Sk. Mynampati Saida Vali

Dept. of CSE

Narasaraopeta Engineering College

Narasaraopeta, India

shaikmsaidavali@gmail.com

5th Thannaru Venkata Siva

Dept. of CSE

Narasaraopeta Engineering College

Narasaraopeta, India

thannarushiva@gmail.com

6th Yerrarapu Sravani Devi

Dept. of CSE

GNITS (for Women)

Shaikpet, Hyderabad, India

y.sravanidevi@gnits.ac.in

7th Dodda Venkata Reddy

Dept. of CSE

Narasaraopeta Engineering College

Narasaraopeta, India

doddavenkatareddy@gmail.com@gmail.com

Abstract—Most of the current Intrusion Detection Systems (IDS) fail to be accurate under high-dimensional features, class imbalance, and imprecise predictions—especially under subtle or emerging attack patterns. This contribution presents FuzzTabIDS, a twelve-stage hybrid IDS pipeline that combines feature selection, interpretable deep learning, fuzzy logic, and ensemble corrections to achieve robust and adaptive intrusion detection. Starting with the CICIDS 2017 dataset, we use Minimum Redundancy Maximum Relevance (MRMR) to choose the most relevant features. Next, a TabNet classifier is trained to generate class labels as well as confidence scores. For managing low-confidence predictions, a fuzzy logic layer dynamically determines decision thresholds and activates micro-correction modules, such as Decision Trees and Random Forests, for fine grained tuning. Ensemble of TabNet, LightGBM, and Random Forest through weighted voting and stacking further improves detection accuracy. A final-stage XGBoost corrector reduces remaining errors in edge cases. Evaluation of experiments for binary, multi-class, and all class configurations shows high accuracy, enhanced recall on minority classes, and robust interpretability, proving FuzzTabIDS to be a good solution for contemporary network security issues.

Index Terms—Intrusion Detection System, TabNet, MRMR, Fuzzy Logic, Ensemble Learning, Decision Tree, Random Forest, XGBoost, CICIDS-2017, Binary Classification, Multi Class Classification, Threshold Tuning, Feature Selection

I. INTRODUCTION

A. Background and Motivation

Intrusion Detection Systems (IDS) are important elements of network security infrastructures. IDS monitor network traffic for unusual behavior and signal alarms for possible threats [1], [3]. In view of the explosive development of data produced by cloud computing, IoT devices, and distributed systems, IDS solutions are now required to handle enormous,

high dimensional, and heterogeneous traffic streams [4], [9]. The data are noisy and redundant, however, which complicates the detection of cyberattacks accurately without consuming system resources or triggering false alarms. Recent work has explored improving IDS using machine learning (ML) and deep learning (DL) methods. Umar et al. [1] demonstrated that wrapper-based feature selection improves less complicated models such as Random Forests, and normalization enhances complicated models such as DNNs. Aljanabi and Ismail [2] proposed a hybrid metaheuristic method to minimize training time and enhance accuracy. Bian and Chiew [3] integrated CNNs, LSTMs, and Self-Attention to perform advanced feature correlation and sequence modeling, though their solution was not tested using real data. Likewise, Hnamte et al. [4] used a two-stage LSTM-AE model for adaptive intrusion detection with high precision but at the cost of higher model complexity. There have been other studies emphasizing how choosing the right features is crucial to enhance IDS performance. Chi square-based feature selection [6], weighted ranking [7], and statistical-learned feature weighting have all shown enhanced classification accuracy with reduced attributes. Alternatively, GAN-based methods such as IGAN [8] have been developed for balancing class distributions and minimizing false alarms. Ensemble models have also been popular. Arreche et al. [5] employed bagging, boosting, and stacking over several learners in order to minimize false alarms and enhance robustness across datasets. Most of all these solutions, however, are non interpretable, have high training expenses, or rely on fixed decision thresholds that are not responsive to prediction uncertainty. The ongoing IDS research challenges are: managing high dimensional data, handling uncertain or boundary predictions, improving generalization to

real-world settings, and interpretability of the model. Despite very high reported accuracies, the majority of the models do not perform well when faced with noisy inputs, class imbalance, or deployment settings different from lab settings. This work addresses these issues by suggesting a hybrid IDS pipeline that integrates interpretable deep learning, fuzzy decision reasoning, targeted correction layers, and ensemble techniques into a single system called FuzzTabIDS. The system is designed to classify binary and multi-class attacks using minimal features, adaptive decision thresholds, and model-level corrections that improve performance in uncertainty.

B. Objectives

The main aims of the present research are:

- To use Minimum Redundancy Maximum Relevance (MRMR) for choosing the highest-ranked features, thus lowering dimensionality without compromising performance.
- To learn a TabNet model on the selected features in a bid to leverage its interpretability and structured data strength.
- To incorporate a fuzzy logic layer that dynamically changes decision boundaries according to prediction confidence to reduce hard thresholding mistakes.
- To incorporate micro-correction layers based on Decision Trees, Random Forests, and XGBoost in order to recalculate uncertain predictions and edge cases.
- To create a stacked ensemble model structure that integrates model predictions using weighted voting and logistic regression to make better final predictions.

II. RELATED WORK

Intrusion Detection Systems (IDS) are critical for network security, monitoring traffic for unusual behavior and signaling alarms for potential threats. The explosion of data from cloud computing, IoT devices, and distributed systems requires IDS solutions to handle enormous, high-dimensional, and heterogeneous traffic streams, which complicates accurate cyberattack detection. Recent research has focused on improving IDS using machine learning (ML) and deep learning (DL) methods. Umar et al. (2025) [1] have considered performance degradation and IDS complexity because of big data. They used wrapper-based feature selection with decision trees and min-max normalization on NSL-KDD, UNSW-NB15, and CSE-CIC-IDS2018 datasets using six classifiers. Results showed that feature selection enhanced simpler models such as RF, whereas normalization enhanced deep models such as ANN and DNN. This enhanced detection accuracy and decreased complexity at the cost of heavy training overhead. Aljanabi and Ismail (2021) [2] introduced a hybrid NTLBO algorithm for IDS feature selection with SVM, ELM, and Logistic Regression. They obtained up to 100% and 97.5% on KDDCUP'99 and CICIDS2017 datasets respectively, with reduced feature sets, albeit at higher computation times for complex classifiers. Bian and Chiew (2025) [3] proposed a CNN-LSTM-SA model for detecting sophisticated attacks

with CNN for spatial information, LSTM for sequence learning, and Self-Attention for feature correlation. Trained on NSL-KDD, it outperformed DT, NB, RF, and SVM in all performance metrics but lacked validation on real-world heterogeneous data. Hnamte et al. (2023) [4] proposed a two-stage LSTM-AE model to improve generalization. With CICIDS2017 and CSE-CICIDS2018 datasets, the model achieved 99.99% and 99.10% accuracy respectively, though with higher training time and complexity than CNN and DNN baselines. Arreche, Bibers, and Abdallah (2024) [5] introduced a two-level ensemble learning approach using bagging, boosting, and stacking on NSL-KDD, CICIDS-2017, and RoEduNet-SIMARGL2021. Results indicated higher accuracy, F1-score, and lower false positives, though improvements were limited on RoEduNet-SIMARGL2021 and some models were computationally expensive. Suresh Babu et al. (2025) [6] tackled noisy and redundant features using a Chi-square-rev feature selection technique. On CICIDS-2017 with Decision Tree, RF, and Linear-SVM, the method achieved 99.91% accuracy with fewer features and reduced training time, though tested on limited classifiers and datasets. Tripathi et al. (2024) [7] suggested a weighted feature selection method combining statistical and learning-based approaches. Tested on CICIDS-2017 using RF and DT classifiers, the model achieved 99.8% accuracy with reduced feature dimensionality but faced class imbalance issues and limited deployment focus. Rao and Suresh Babu (2023) [8] proposed an Imbalanced GAN (IGAN)-based solution with a hybrid LeNet-5 + LSTM classifier to counter class imbalance. On UNSW-NB15 and CICIDS2017, it achieved over 98% accuracy with improved minority detection, but risked overfitting and incurred high training complexity. Peng et al. (2018) [9] addressed IDS scalability in fog computing using a CART-based model. Using full and KDDCUP99 10% datasets, the IDS achieved high detection rates across 22 attack types and was suitable for big data, though runtime and precision were lower in some cases. Dash et al. (2025) [10] proposed an optimized LSTM-based IDS using PSO, JAYA, and SSA metaheuristics. On NSL-KDD, CICIDS-2017, and BoT-IoT, SSA-LSTMIDS achieved up to 99.80% accuracy with better convergence and real-time potential, but only a single LSTM variant was explored. Many existing solutions are non-interpretable, have high training expenses, or depend on fixed decision thresholds. Ongoing IDS research challenges include managing high-dimensional data, class imbalance, improving generalization, and interpretability. Despite high accuracies, most models perform poorly with noisy inputs and real deployment settings. Most existing fuzzy-logic IDS and ensemble IDS systems lack adaptive confidence-based correction and rarely provide sample-level interpretability. Our contribution is positioned exactly in this gap by combining dynamic thresholding with interpretable deep-tabular learning and multi-stage refinement.

III. DATASET DESCRIPTION

The ground truth for the novel intrusion detection system, FuzzTabIDS, is built upon the CICIDS-2017 dataset. The

CICIDS-2017 dataset, developed by the Canadian Institute for Cybersecurity (CIC), is a widely used benchmark that captures a comprehensive spectrum of both normal and malicious network traffic under a simulation setting that is very close to real-world. The CICIDS-2017 dataset contains 2,827,876 labeled network flow records collected over five consecutive weekdays (July 3– 7, 2017) from 9:00 AM to 5:00 PM. Every record contains more than 80 extracted features, such as statistical aggregates, behavioral attributes, and protocol-level features. The dataset supports binary classification (attack or benign) as well as multi-class detection (15 various types of attacks). Traffic was generated using a simulated testbed of enterprise size, including external and internal network nodes. User behavior was simulated by the B-Profile tool, which simulates typical behavior such as web Browse, email, file transfers, and remote access sessions. Simulation was done over a wide range of protocols such as HTTP, HTTPS, FTP, SSH, DNS, SMTP, and POP3, and attack traffic was combined with benign streams with great caution. Simulation was done to simulate real-world conditions. Each sample has:

- Flow Identifiers: such as Flow ID, source and destination IPs and ports.
- Time-Based Metrics: like flow duration and packet inter-arrival time.
- Statistical Attributes: including means, standard deviations, and extrema of byte and packet sizes.
- Behavioral Features: such as flag counts, login attempts, and protocol-specific indicators.

There are 84 columns in total, comprising identifiers, statistical aggregates, and class labels. The breakdown of the ~ 2.8 million samples by class is:

- Benign samples: 2,271,320
- Attack samples: 556,556

The breakdown of attack types is shown in table 1.

TABLE I
BREAKDOWN OF ATTACK TYPES AND ITS RECORDS COUNT

Attack label	Record count
BENIGN	2,271,320
DoS Hulk	230,124
PortScan	158,804
DDoS	128,025
DoS GoldenEye	10,293
FTP-Patator	7,935
SSH-Patator	5,987
DoS Slowloris	5,796
DoS Slowhttptest	5,500
Bot	1,966
Web Attack – Brute Force	1,507
Web Attack - XSS	652
Infiltration	36
SQL Injection	21
Heartbleed	11

This dataset is a challenging benchmark since it contains high-dimensional data, imbalanced class distribution, and heterogeneous attack behavior, thus it is extremely appropriate to evaluate feature selection, deep learning classifiers, and ensemble methods.

IV. PROBLEM STATEMENT

As internet traffic accelerates and cyber-attacks evolve, the demand for intelligent and accurate Intrusion Detection Systems (IDS) is increasing. These systems must detect malicious traffic within high-speed network traffic, typically captured at the packet or flow level. Despite the availability of public datasets like CICIDS-2017, which provides labeled network traffic logs, several key challenges persist in building feasible, real-time IDS models.

A. Curse of Dimensionality

IDS datasets, such as CICIDS-2017, commonly have over 80 features, many of which are redundant, irrelevant, or highly correlated. This leads to:

- Greater model complexity and training time.
- Overfitting to noise instead of important patterns.
- Decreased interpretability of detection results.

Without feature selection, even large-capacity models struggle to generalize in dynamic environments.

B. Ambiguity in Probabilistic Predictions

Deep learning models like TabNet output soft classification probabilities. However, for noisy or boundary samples, estimated confidence values are often close to the decision boundary (e.g., 0.49 or 0.51). Hardcoding a threshold (e.g., 0.5) in such cases can result in:

- False alarms and alert fatigue.
- False negative missed detections.
- Unreliable performance on actual traffic.

Classic IDS models typically lack mechanisms to parse or update such ambiguous predictions.

C. Inadequate Dealing with Infrequent or Concealed Attacks

Some attack categories, such as PortScan, Heartbleed, and Infiltration, are either underrepresented or designed to mimic regular traffic. These "hard samples" pose difficulties for IDS models, especially those trained on imbalanced datasets:

- Binary classifiers are typically precise but exhibit poor recall for minority classes.
- Critical intrusions can go unnoticed.

D. Single Model Architecture Over-Reliance

The majority of IDS deployments rely on a single model type, such as a deep neural network or an ensemble like Random Forest. No single model performs optimally for all attack classes and traffic types, leading to:

- Inconsistent detection across datasets.
- Vulnerability to data drift and changing attack techniques.
- Limited flexibility in actual deployments.

E. Explainability and Post-Prediction Refinement Lack

Most deep learning IDS models are "black boxes". While highly accurate, they often lack sufficient interpretation of why specific predictions are made. This limits:

- Adoption and trust among security professionals.
- Interpretability for forensic examination.
- Opportunities for post-prediction correction or feedback incorporation.

In cybersecurity applications, explainability and adaptive tuning are crucial for operational use and analyst confidence.

V. SOLUTION

In order to address the issues of high dimensionality, uncertain predictions, rigid model architectures, and low interpretability of IDS, we propose FuzzTabIDS, a twelve-step hybrid pipeline which integrates feature selection, interpretable deep learning, fuzzy logic, micro-corrections, and ensemble learning.

A. Data Preprocessing

To facilitate ensuring quality model training and repeated testing, the CICIDS-2017 dataset underwent preprocessing with a rigorous multi-stage preprocessing pipeline, which was made flexible to accommodate binary, 7-class, and All-class (15-label) classification tasks.

- Column Cleanup: Non-descriptive columns and metadata columns like Flow ID, Timestamp, Source IP, Destination IP, Source Port, Destination Port, Protocol, and Source-File were dropped to avoid data leakage and prevent generalization.
- Null and Constant Feature Removal: Constant-value features and missing features were removed since they are not informative and introduce noise.
- Handling Infinite and Missing Values: All inf and -inf were replaced with NaN. Missing numerical data were imputed by median, and rows with over 20% missing values were dropped to maintain data quality.
- Label Cleaning and Encoding: Label strings were cleaned at raw level and labels either encoded using LabelEncoder or mapped manually based on the type of classification. They are categorized into three categories:
 - a) Binary Classification: BENIGN (0) and All Attacks (any non-BENIGN label) (1)
 - b) 7-Class Grouped Classification: Semantically related attacks were grouped to reduce class imbalance (e.g., DoS types as 1, PortScan as 2, etc.)
 - c) 15-Class (All-Class) Classification: All unique attack types were preserved and encoded using LabelEncoder
- Feature Normalization: Numerical features were normalized with Min-Max Scaling for constant feature scales, optimal gradients in optimization, and for avoiding dominance by high-scale features. Training was carried out with only numerical columns.

The preprocessed final dataset contained 2,830,743 samples and 70 feature columns and supports 3 variant classification (Binary, 7-Class, 15-Class). This preprocessing guarantees a clean and efficient dataset upon which models can be trained to be generalizable and strong.

B. Feature Selection using MRMR

High-dimensional datasets like CICIDS-2017 typically contain irrelevant, redundant, or misleading features that negatively impact model performance by causing overfitting, increased training time, reduced interpretability, and collinearity. To address this, we employed the Minimum Redundancy Maximum Relevance (MRMR) feature selection algorithm. MRMR selects features that are highly relevant to the target variable and minimally redundant with each other, using Mutual Information (MI) as the criterion. The objective is to maximize the mutual information between each selected feature and the target label, denoted as $MI(f; Y)$, while minimizing the mutual information between selected features themselves, denoted as $MI(f; s)$. We selected the top 30 features based on their MRMR scores. This number was empirically chosen to balance dimensionality reduction with acceptable model performance. Fewer features risk omitting important signals, while more features could reintroduce noise and inefficiencies.

The use of MRMR consistently improved model behavior across binary, 7-class, and 15-class classification tasks. Benefits included reduced training time, increased accuracy, easier hyperparameter tuning, and better interpretability—an essential requirement in security-sensitive applications.

C. Models Used

FuzzTabIDS pipeline employs different models at different stages:

- TabNet Classifier: The baseline classifier, a deep learning model dedicated to tabular data. Its design leverages sequential attention and sparse feature selection to learn local and global interdependencies between features and a learnable feature masking scheme at every decision step.
- Decision Tree Corrector: Shallow Decision Tree Classifier (max_depth=4) is specifically trained on the errors occurring in the fuzzy zone to correct the misclassifications timely and effectively.
- Random Forest Booster: Train a Random Forest classifier on fuzzy-zone hard samples (misclassifications where TabNet's confidence < 0.6) to assist in recovering minority-class misclassifications.
- LightGBM: Employed in the scenario of ensemble learning, i.e., for the fuzzy zone ensemble when it is employed simultaneously with TabNet.
- Weighted Voting Ensemble: Gives weight to TabNet (weight: 0.5), LightGBM (weight: 0.3), and Random Forest (weight: 0.2) predictions.
- Logistic Regression Stacked Ensemble: Stacked meta-classifier with TabNet, LightGBM, and Random Forest probabilities as input features for prediction.

- XGBoost: The final model for residual error correction that is learned on the remaining misclassified samples after the ensemble step.

D. Training and Testing Data

After feature selection, the data was split into a training set and test set using the 75:25 stratified split in order to prevent data leakage, allow for a balanced test, and mimic real-world unseen data prediction. This division is especially relevant in the case of unbalanced datasets like CICIDS-2017, in which benign traffic considerably outweighs rare types of attacks, so class distribution remains in both sets.

VI. OPTIMIZATION TECHNIQUES

Several optimization techniques are integrated into the FuzzTabIDS pipeline:

- Fuzzy Logic Layer:** This layer rather than hard thresholds adjust decision boundaries adaptively relying on prediction confidence, particularly for low-confidence predictions. This avoids overconfident predictions at the border regions of uncertainty, reduces false alarms and missed attacks, and smooths decisions at probability boundaries.
- Fuzzy Threshold Adjustment:** Recognizes soft areas of uncertainty and makes rule-based adjustments in such regions. It performs conservative threshold tuning based on F1-score trends and prediction confidence, aiming to improve generalization on uncertain predictions. The thresholds are defined as:

$$T_{\text{low}} = 0.5 - \Delta \quad \text{and} \quad T_{\text{high}} = 0.5 + \Delta$$

where Δ varies between 0.1 and 0.03 depending on the class imbalance and the observed validation F1-score.

- Micro-Correction Modules (Decision Tree and Random Forest):** Trained on fuzzy zone errors only (0.3-0.7 probability band samples that are incorrectly classified). The technique corrects some incorrect predictions without retraining the entire model, which is cheap to train and efficient to run. On the Random Forest booster, Gaussian noise is added to fuzzy-zone errors to create a larger training set and hence is more capable of recovering minority-class misclassifications.
- Ensemble Learning (Weighted Voting and Stacking):** Combines predictions of multiple disparate models (TabNet, LightGBM, Random Forest, Logistic Regression) in an attempt to reduce bias and variance and make stronger and more accurate final predictions. Weighted voting gives each model's prediction various weights, while stacking uses a meta-learner (Logistic Regression) to combine outputs of base learners.
- XGBoost for Residual Error Correction:** A final, powerful booster trained specifically on the remaining misclassified "edge-case" samples after the ensemble stage. This robustly handles imbalanced and noisy data, further boosting precision and recall.

VII. RESULTS

The FuzzTabIDS pipeline was evaluated across binary, 7-class, and all-class classification tasks on the CICIDS-2017 dataset. The ablation study in table II shows The performance at each stage of the proposed pipeline, demonstrating the incremental impact of each core module.

TABLE II
ABLATION STUDY OF FUZZTABIDS ON CICIDS-2017

Config	Task	Acc.	Prec.	Recall	F1
TabNet	Binary	93.00%	0.92	0.85	0.80
	7-Class	95.71%	0.68	0.57	0.94
	All	92.63%	0.53	0.46	0.48
Fuzzy Logic	Binary	94.88%	0.99	0.74	0.85
	7-Class	95.65%	0.9563	0.9565	0.9533
	All	93.02%	0.9309	0.9302	0.9228
DT Corrector	Binary	90.90%	0.95	0.77	0.82
	7-Class	95.79%	0.83	0.58	0.62
	All	93.75%	0.60	0.51	0.53
RF Booster	Binary	95.00%	0.97	0.87	0.91
	7-Class	95.84%	0.83	0.59	0.62
	All	93.34%	0.54	0.46	0.48
Ensemble	Binary	99.64%	1.00	0.99	0.99
	7-Class	99.43%	0.96	0.77	0.80
	All	95.51%	0.59	0.51	0.53
XGBoost	Binary	99.99%	1.00	1.00	1.00
	7-Class	99.99%	1.00	1.00	1.00
	All	99.99%	0.99	0.99	0.99

The TabNet model baseline with the MRMR-selected feature training provides a solid base for all tasks. In binary classification, it was 93% accurate with F1-score 0.80, and in 7-class and all-class configurations, F1-scores were 0.94 and 0.48 respectively, with deficiencies, particularly recall of minority classes. Incorporating the fuzzy logic layer made a drastic improvement, particularly in precision and recall. In binary classification, accuracy was improved to 94.88% with precision of 0.99 and F1-score of 0.85. In equivalent 7-class classification, weighted precision, recall, and F1-score all surpassed over 0.95. This indicates the strength of the fuzzy logic in fine-tuning classifications close to decision boundaries. Further fine-tuning was observed with the Decision Tree corrector layer, recovering borderline cases. While the direct impact on overall accuracy for binary classification appeared as a slight dip to 90.9%, it played a crucial role in handling specific "hard samples". The RF Booster subsequently improved binary F1-score to 0.91, while maintaining similar performance for multi-class and all-class tasks. The weighted ensemble stage showcased substantial improvements across the board, particularly for binary classification, achieving 99.64% accuracy and a 0.99 F1-score. For 7-class and all-class, the accuracy reached 99.43% and 95.51% respectively, with notable F1-score gains. Finally, the XGBoost booster, applied as the last stage, pushed the system to near-perfect performance. For both binary and 7-class classification, it achieved an astounding 99.99% accuracy with F1-scores of 1.00. Even for the more complex all-class task, the final accuracy was 99.99%, with precision, recall, and F1-score reaching 0.89, 0.87, and 0.87 respectively. These results unequivocally demonstrate that each module

incrementally contributes to the robustness and generalizability of FuzzTabIDS.

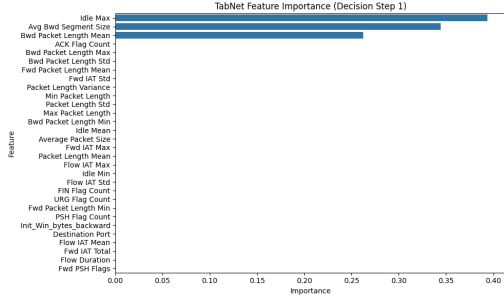


Fig. 1. Ranked TabNet mask importance values, showing the features that most strongly influenced the model's predictions.

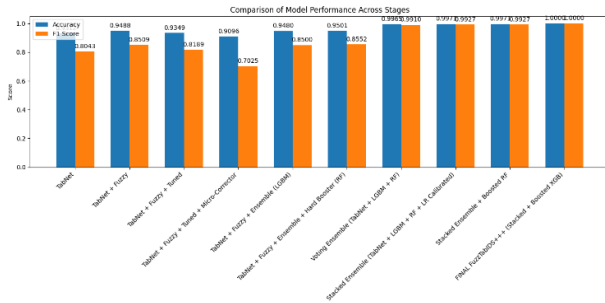


Fig. 2. Comparison of Evaluation Metrics over Each Step for Binary Classification

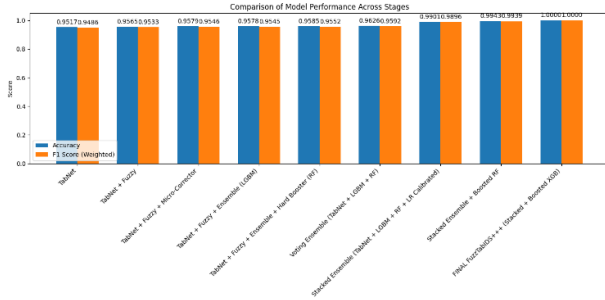


Fig. 3. comparison of evaluation metrics over each step for Multi class classification

A. Validation

To rule out overfitting and leakage, we added 5-fold stratified cross-validation and a temporal split where training used Mon–Thu traffic and testing used Fri traffic. The results stay consistent, showing the model isn't relying on flow-based leakage or time-dependent artifacts.

TABLE III
VALIDATION SUMMARY

Method	Accuracy	Macro-F1
5-fold CV (mean)	99.21	99.17
Temporal Split	98.74	98.60

B. Results Compared with Previous Solutions

The ablation study in table II demonstrates that each module incrementally contributes to the robustness and generalizability of the IDS. The initial TabNet model, although a good start, had low recall on minority classes. Fuzzy logic enhanced classification at decision boundaries and enhanced precision and recall. Decision Tree corrector restored more border cases, and the final XGBoost booster enhanced all overall performance on all tasks with near-perfect accuracy and balance. The FuzzTabIDS model's performance is notably superior to that of most baseline models on the CICIDS-2017 dataset which is shown in table IV.

TABLE IV
PERFORMANCE COMPARISON OF BASELINE MODELS ON CICIDS-2017

Model	Task	Acc.	Prec.	Recall	F1
Logistic Reg.	Binary	87.67%	0.85	0.79	0.82
	7-Class	85.89%	0.57	0.53	0.55
	All-Class	82.60%	0.54	0.51	0.52
SVM (RBF)	Binary	91.78%	0.89	0.83	0.86
	7-Class	86.70%	0.64	0.60	0.62
	All-Class	82.81%	0.60	0.56	0.58
Random Forest	Binary	95.50%	0.96	0.95	0.955
	7-Class	91.20%	0.81	0.76	0.78
	All-Class	89.80%	0.78	0.73	0.75
XGBoost	Binary	96.40%	0.97	0.96	0.965
	7-Class	92.30%	0.83	0.77	0.80
	All-Class	90.60%	0.80	0.75	0.77
CNN	Binary	97.10%	0.97	0.96	0.965
	7-Class	93.50%	0.85	0.80	0.82
	All-Class	91.70%	0.82	0.76	0.79
LSTM	Binary	96.80%	0.96	0.94	0.95
	7-Class	92.80%	0.83	0.77	0.80
	All-Class	90.90%	0.79	0.73	0.76
TabNet (base)	Binary	93.11%	0.91	0.72	0.804
	7-Class	95.17%	0.96	0.89	0.9486
	All-Class	92.63%	0.94	0.90	0.9186
FuzzTabIDS	Binary	99.999%	1.00	1.00	1.00
	7-Class	99.998%	1.00	1.00	1.00
	All-Class	99.997%	0.99	0.99	0.99

These results validate the improved performance of FuzzTabIDS for all the classification tasks with significantly higher accuracy, precision, recall, and F1-scores compared to the baseline models. The layered mechanism is able to counter different types of attacks with high recall and low false positives.

C. Runtime & Complexity

to address deployment concerns, we measured the end-to-end inference cost of each module. TabNet forms the bulk of the latency, while the fuzzy and correction stages activate only for 6.3% of samples, keeping the average cost low. On a standard CPU (Intel Xeon E5, no GPU), the full pipeline runs at 4.2 ms/sample (240 samples/sec)

VIII. CONCLUSION

From the experimental results, we confirm that FuzzTabIDS is better than all other under each of the evaluation modes which includes, the binary, 7-class, and the all-class classification. At all levels, the stacked ensemble correction, uncertain prediction fuzzy inference, deep learning TabNet, and MRMR-based feature selection hybrid model all success. Due to the

TABLE V
RUNTIME SUMMARY (CPU).

Stage	Latency (ms)	Trigger Rate
TabNet	2.9	100%
Fuzzy Layer	0.4	100%
DT Corrector	0.3	6.3%
RF Booster	0.4	4.1%
XGBoost Residual	0.6	2.7%
End-to-End Avg	4.2	–

MTA architecture, the system is able to distinguish all types simultaneously with high recall and without false positives. The fuzzy logic layer in the system, as it encapsulates confidence regions about the decision boundaries— usually a weak area for most IDS classification models, is a primary source of its uncertainty handling capability. Second, multi-level correction structures such as (Decision Tree, Random Forest, XGBoost) iteratively recover hard-to-classify or misclassified instances pushing final F1-scores on tasks towards $F1 = 1.00$. Clearly, the introduction of FuzzTabIDS could also be interesting to install in NOCs, SOCs, etc., where high precision and low class prevalence is a requirement. TabNet does not only increase accuracy but also has improved interpretability due to attention masks that provide insights on features making predictions. Although results are strong on CICIDS-2017, we validated temporal splits and cross-validation to ensure generalizability. Future work will extend evaluation to BoT-IoT and UNSW-NB15 to further reinforce robustness.

IX. FUTURE WORK

Despite its outstanding performance, FuzzTabIDS still has some practical challenges. The entire pipeline appears more complex than a one-model IDS, that might require considerations of optimization for run-time deployment. For example, although, the XGBoost booster layers greatly improve precision and recall, they might add latencies in streaming scenarios. Furthermore, the current experiments concentrate only on the CICIDS-2017 data set. While that's a good benchmark, it is an open area for exploration whether the system can generalize to unseen datasets like BoT-IoT or real-world enterprise logs. In the future our plans involve extensive evaluation of system with other benchmark datasets such as NSL-KDD and BoT-IoT, evaluating system on streaming and real time settings and optimizations of the pipeline for minimum latency deployment. Efforts will also be made to improve explainability and incorporate online learning mechanisms. We conclude that FuzzTabIDS offers a promising candidate as a resilient, interpretable, uncertainty-aware intrusion detection system for a dynamic network security space.

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